

POSTLAB

AIM:-

Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.

CODE:-

```
GNU nano 8.7 postl
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, f, i, j, k, flag, page_faults = 0, index = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    // Initialize frames
    for(i = 0; i < f; i++) {
        frames[i] = -1;
    }

    printf("\nPage\tFrames\n");

    for(i = 0; i < n; i++) {
        flag = 0;

        // check if page is already in frame
        for(j = 0; j < f; j++) {
            if(frames[j] == pages[i]) {
                flag = 1;
                break;
            }
        }
    }

    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Exe
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Jus
```

```
GNU nano 8.7 postlab.c

for(i = 0; i < n; i++) {
    flag = 0;

    // check if page is already in frame
    for(j = 0; j < f; j++) {
        if(frames[j] == pages[i]) {
            flag = 1;
            break;
        }
    }

    // If page not found → page fault
    if(flag == 0) {
        frames[index] = pages[i];
        index = (index + 1) % f;
        page_faults++;
    }

    // Print frames
    printf("%d\t", pages[i]);
    for(k = 0; k < f; k++) {
        if(frames[k] != -1)
            printf("%d ", frames[k]);
        else
            printf("- ");
    }
    printf("\n");
}

printf("\nTotal Page Faults = %d\n", page_faults);

return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Ex
^X Exit ^R Read File ^\ Replace ^U Paste ^J Ju

OUTPUT:-

```
Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ nano postlab.c

Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ gcc postlab.c -o postlab

Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ ./postlab
Enter number of pages: 8
Enter page reference string:
1 2 3 4 1 2 5 1
Enter number of frames: 3

Page      Frames
1          1 - -
2          1 2 -
3          1 2 3
4          4 2 3
1          4 1 3
2          4 1 2
5          5 1 2
1          5 1 2

Total Page Faults = 7

Nandini Kasare@LAPTOP-NIHKGM4O MINGW64 ~
$ |
```